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Composition of the Pheromone System of the male Danaine Butterfly, *Idea leuconoe*

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Abstract

Male *Idea leuconoe* butterflies release a complex mixture of volatiles from their pheromone glands (hairpencils) during courtship. The pheromone components geranyl methyl thioether (**2**), viridifloric-β-lactone (**3**), and 6-hydroxy-4-dodecanolide (**10**) have been synthesized for the first time. Therefore the structural assignment of these new natural products could be proved. Related 7-hydroxy-5-alkanolides are also present in the extract. The volatiles are embedded in a lipidic matrix with more than 150 components. This matrix consists of alkanes, alkenes, 2,5-dialkyltetrahydrofurans, secondary alkanols and alkenols as well as alkanones and alkenones. Several regioisomers of the oxidized hydrocarbons occur. The elucidation of double bond positions has been performed by MS using DMDS adducts.

Introduction

Male Danaine butterflies possess striking evertible pheromone glands, so called hairpencils, which are used during courtship [1]. About 30 years ago the first pheromone component of these butterflies, danaidone (**1**), could be identified [2] and its function as courtship pheromone proved [3]. Since then, many danaine species have been investigated and the chemical composition of their male pheromone glands elucidated [1, 4, 5]. Despite the fact that some of the scent bouquets consist of up to 60 components, no pheromonal function of any other component than **1** has been established.

Recently we were able to show that male *Idea leuconoe* butterflies emit a complex mixture of chemicals from their hairpencils. At least three of these components, danaidone (**1**), geranyl methyl thioether (**2**), and (*S,S*)-viridifloric-β-lactone (**3**), act as courtship pheromones [6]. An artificial mixture of hairpencil compounds containing **1**, **2**, and **3** as well as phenol, *p*-cresol, benzoic acid, a series of homologue 6-hydroxy-4-alkanolides ranging from C_{10} to C_{13}, (*E,E*)-farnesol, and (*Z*)-9-tricosene elicited the same courtship behavior as a crude hairpencil extract [6]. In addition, (–)-(*R*)-mellein (8-hydroxy-3-methyl-3,4-dihydroisocoumarin) and another β-lactone related to **3**, 2-ethyl-2-hydroxy-3-butanolide, could be identified in the hairpencil extracts. Besides their function as courtship pheromones, other types of interactions, like a defensive warning odor or intermale recognition (for a discussion, see [6]), seem also to be associated with these chemicals. The volatile compounds identified are embedded into a complex lipidic matrix (see Figure 1). In this paper we will report on the synthesis of some hairpencils constituents, the identification of additional components, and the composition of the lipid matrix. fig. 1 here

Figure 1: Figure

Results

The structure of the previously unknown geranyl methyl thioether (**2**) could be deduced from its mass spectrum (see Figure **??**). The presence of sulfur was detected by high-resolution mass spectroscopy, giving a molecular formula of $C_{11}H_{20}S$ (M^+_{obs} = 184.1289, M^+_{calc} = 184.1286). Ions at m/z = 47 (CH_3S^+) and 61 ($CH_3SCH_2^+$) and the loss of 48 amu (CH_3SH) from M^+ indicated a thiomethyl group in the molecule. Typical terpenic ions (m/z = 41, 69, 81, 93, 123, and 136) suggested that this molecule could be geranyl methyl thioether (**2**). For an unambiguous proof, **2** was synthesized by reaction of geranyl chloride with sodium methanolate in boiling ethanol. The product showed identical mass spectra and gaschromatographic retention times on different stationary phases as the natural compound which is therefore (*E*)-2,6-dimethyloctadienyl methyl thioether (**2**). The corresponding (*Z*)-isomer exhibited a similar mass spectrum, but a shorter retention time on an apolar phase. fig. **??** here

Another pheromone component not reported from nature before is 2-hydroxy-2-(1-methylethyl)-3-butanolide (viridifloric β-lactone, **3**). Its identification basing on MS-, IR- and NMR-data has been described [6]. A racemic mixture of both diastereomers of **3** was synthesized for structural assignment according to Figure 4.

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Compound numbers

1 danaidone
2 geranyl methyl thioether
3 viridifloric acid lactone
4 ho but sre
5 butensre
6 virid. acid
7 oxobutester
8 hexenolid
9 oxo olid
10 6-ho-c12-g-lactone